MINOR PROJECT SYNOPSIS

INSTA MECHANIC

On-Demand Vehicle Repair and Service Platform

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Submitted To

Department of Computer Science & Engineering

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Dehradun, Uttarakhand

Academic Session: 2025 – 2026

Introduction

Insta Mechanic is a modern web-based platform developed to provide instant vehicle repair and maintenance services during emergency and breakdown situations. The platform connects users with nearby verified mechanics through a location-based system. It is designed using HTML, CSS, Bootstrap, JavaScript, Node.js, Express.js, and PostgreSQL/Supabase technologies.

Problem Statement

Vehicle breakdowns create inconvenience and safety concerns for users. Traditional methods of finding mechanics are time-consuming and unreliable. Insta Mechanic solves this problem by offering real-time mechanic booking and verified repair services.

Objectives of the Project

- Develop a responsive web platform for mechanic services.
- Provide real-time access to nearby mechanics.
- Implement secure authentication.
- Create dashboards for customers, mechanics, and admins.
- Improve transparency and service efficiency.

Scope of the Project

The platform supports roadside assistance, mechanic verification, booking management, service tracking, and dashboard management. Future scope includes AI recommendations, mobile apps, GPS tracking, and online payment integration.

Literature Review

Research on location-based systems and emergency service platforms highlights the importance of GPS-enabled assistance and verified service providers. Existing systems lack transparency and efficient response mechanisms, which Insta Mechanic aims to improve.

Methodology

The project follows Requirement Analysis, System Design, Frontend Development, Backend Development, Database Integration, Testing, and Deployment phases for structured implementation.

Technologies Used

HTML, CSS, Bootstrap, JavaScript, Node.js, Express.js, PostgreSQL/Supabase, JWT Authentication, Git, and GitHub.

Feasibility Study

The project is technically feasible using open-source technologies, economically feasible with minimal development cost, and operationally feasible due to its user-friendly design.

Expected Outcomes

A secure and reliable mechanic booking platform that reduces response time, improves customer safety, and provides efficient vehicle service management.

Future Enhancements

Future enhancements include mobile applications, live GPS tracking, AI-based recommendations, online payments, and multilingual support.

Conclusion

Insta Mechanic provides a practical and scalable solution for emergency vehicle repair services. The system combines modern technologies with user-friendly design to improve accessibility, reliability, and efficiency in roadside assistance.

References

- Bootstrap Official Documentation
- Node.js Official Documentation
- Express.js Guide
- PostgreSQL Documentation
- Research Papers on GPS-Based Emergency Systems